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RED LIGHT GREEN LIGHT (RLGL)

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This project implements a Red Light, Green Light (RLGL) game using computer vision. A webcam captures real-time video, and the system detects player movement to determine whether the player should continue or lose. The algorithm combines motion detection and a state machine to simulate the game rules. Microsoft 365 Copilot was used to produce the design, algorithm and coding based on the given parameters.

ALGORITHM

The algorithm is built around a continuous video processing loop combined with a finite state machine (FSM). It processes webcam frames in real time to measure motion and make decisions based on predefined thresholds and timing rules.

Steps of the Algorithm

1. Capture a frame from the webcam.
2. Resize the frame and convert it to grayscale.
3. Compute motion score:

motion_score = mean(|current_frame - previous_frame|) / 255

4. Use motion score and timers to update the game state.
5. Display the current state and motion value.
6. Repeat until the game ends or user quits.

STATE MACHINE DESIGN

The game is controlled using a finite state machine (FSM) with four states:

- GREEN – Player must move
- RED – Player must stay still
- WARNING – Player is not moving enough
- DEAD – Game over

State Transitions

GREEN State

- If motion is low → start idle timer
- If idle exceeds warning limit → switch to WARNING
- If idle exceeds death limit → switch to DEAD
- After time expires → switch to RED

WARNING State

- Player is warned for being idle
- If player moves → return to GREEN
- If still idle → DEAD

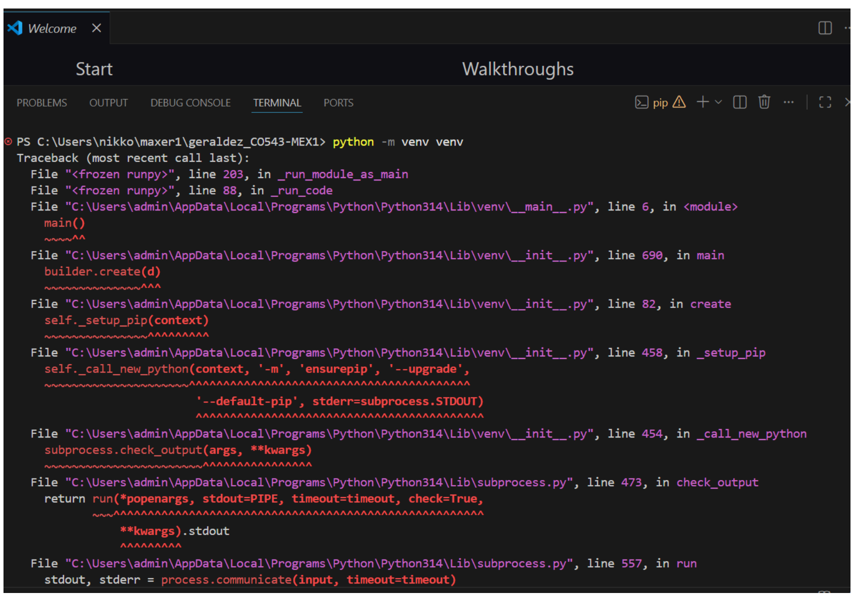
RED State

- Player must stay still
- First few milliseconds = grace period
- After grace:
- If motion is detected → DEAD
- If player survives → return to GREEN

DEAD State

- Game ends and displays “YOU DIED”

ERRORS ENCOUNTERED



This error occurs because the virtual environment creation process was manually interrupted or timed out while downloading and installing pip. The venv module automatically runs ensurepip in the background to bootstrap the pip package manager inside your new environment. This step requires an active internet connection and can sometimes take a long time or hang. The KeyboardInterrupt at the very bottom means you (or a script timeout) cancelled the process (usually by pressing Ctrl+C) while Python was waiting for that installation to finish. To fix the error, the following code was used:

python -m venv myenv --without-pip